

Text as Data: Computational Methods for Social Science Research

How We Can Use Text as Data for Social Science Research

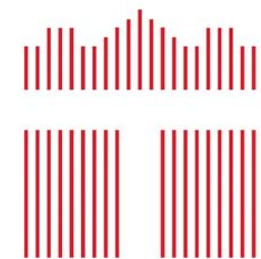
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Part 1 — Text as Data in Political Science

what it is, how the field got here, what it is used for

Why Text?

Most political activity leaves a **textual/digital trace**.

- Parliamentary speeches, hearing transcripts, party manifestos
- Court rulings, bills, administrative records, diplomatic cables
- News coverage, social media posts, politicians' tweets
- Open-ended survey responses, interview transcripts

Then: human reading and hand coding (e.g., the Manifesto Project)

Now: **scale** — millions of documents can be analyzed systematically

What Is “Text as Data”?

Treat text as **data**: turn unstructured language/data into analyzable quantitative information to answer social-science questions.

Quantitative text analysis and **NLP** are the toolkit; the social-science question is the job.

corpus → representation → model → validation → inference

- **Representation**: bag of words, word embeddings, language models
- **Model**: dictionaries, classifiers, topic models, scaling
- **Validation**: comparison with hand coding, expert judgment, external criteria

How the Field Got Here

2003–2010 · Bag of words

Wordscores (2003)

Laver, Benoit & Garry · APSR

Wordfish (2008)

Slapin & Proksch · AJPS

Expressed agenda (2010)

Grimmer · Political Analysis

2013–2014 · Validation

Promise and Pitfalls (2013)

Grimmer & Stewart · Political Analysis
the four principles — and “validate”

Structural topic model (2014)

Roberts et al. · AJPS
topics as a function of covariates

2023– · Meaning in context

Embedding regression (2023)

Rodríguez, Spirling & Stewart · APSR
what a word means, by party and era

LLMs as annotators (2024)

Heseltine & Clemm von Hohenberg
Research & Politics

What each stage added:

bag of words → **validation & covariates** → **meaning in context** → **models that label for you**

Validation is the one demand that never went away.

Sources: Laver et al. (2003); Slapin & Proksch (2008); Grimmer (2010); Grimmer & Stewart (2013); Roberts et al. (2014); Rodríguez et al. (2023); Heseltine & Clemm von Hohenberg (2024)

Roadmap: Two Tasks

Text-as-data research does **two** different things.

Measurement

- How do we turn a concept into a credible text-based measure?
- Contribution: new **method**, new **measure**, new **dataset**
- Star of the show: the method itself
- The textbook (Grimmer, Roberts & Stewart 2022) has three tasks: **discovery**, **measurement**, **inference**.
- Here we collapse them: **measurement** $\approx$ discovery + measurement; **application** $\approx$ inference.

Application

- What substantive question does the measure answer?
- Contribution: new **empirical finding**, theory test
- Star of the show: research design and identification

Example 1: From Pork to Policy

From Pork to Policy: The Rise of Programmatic Campaigning in Japanese Elections

Amy Catalinac, New York University

We examine two related propositions central to the subfield of comparative politics: that candidates for office adopt different electoral strategies under different electoral systems and rely more on particularism when faced with intraparty competition. We apply an innovative methodological approach that combines probabilistic topic modeling with in-depth qualitative interpretations of each topic to an original collection of 7,497 Japanese-language candidate election manifestos used in elections on either side of Japan's 1994 electoral reform. We find that the reform, which eliminated intraparty competition, was associated with a decline in particularism and an increase in promises of programmatic goods such as national security among candidates affiliated with Japan's Liberal Democratic Party. This is not explained by the entry of new candidates or other variables that could plausibly increase discussion of national security. Consistent with the theory, we find that opposition candidates relied on programmatic goods under both electoral systems.

Method: an existing tool, not a new one

LDA (2003), every topic validated by reading.

The contribution is the political science, not a new algorithm.

Building new NLP models is computer science's job; ours is to use them well to answer a question.

Example 2: Open-Ended Answers as Causal Graphs

Jankin, Liao & Tang (2026)

Folk Theories of Partisan Conflict

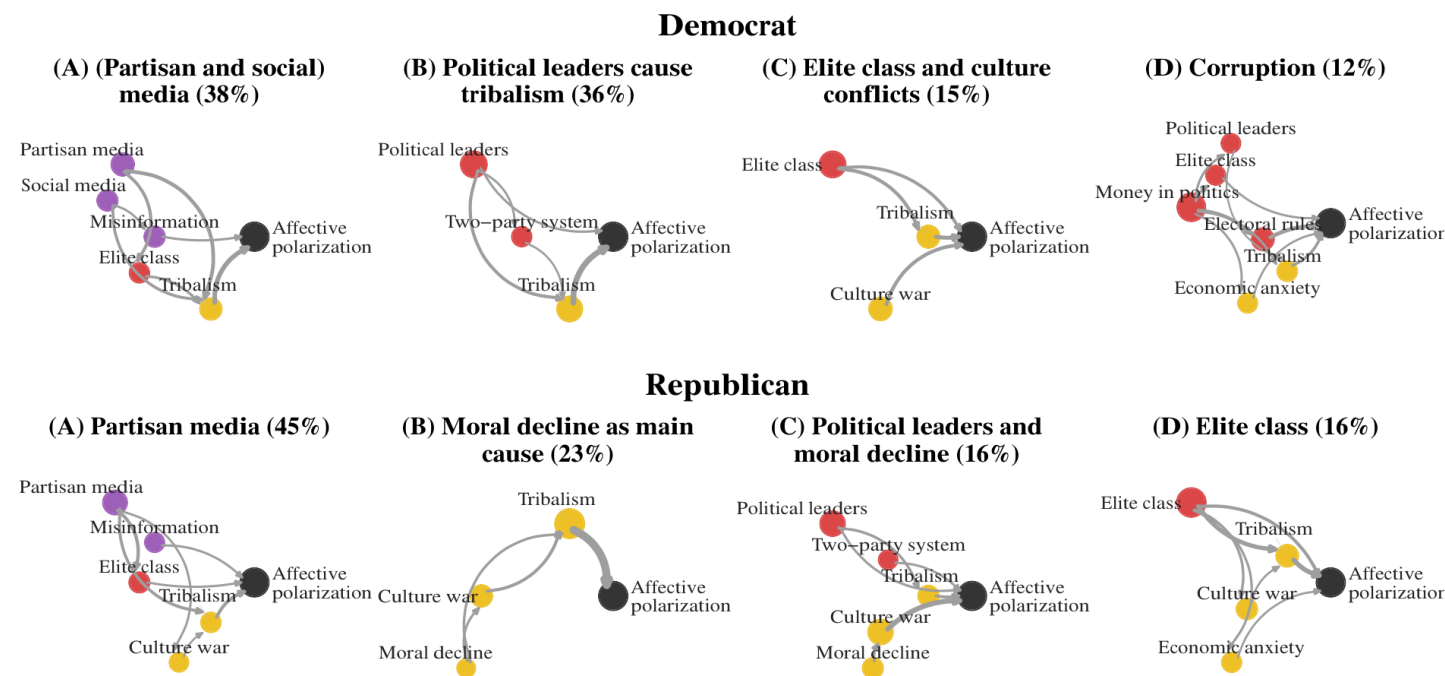
- **Question:** does it matter *why* a partisan thinks the country split — over and above **how much** they dislike the other side?
- **Data:** 1,200 US adults, quota-matched; several open-ended items (the graphs here code Q7)

The measurement move: code each free-text answer as a **directed acyclic graph** — nodes are the factors the respondent names (15 factors, 3 domains), edges are the causal links they draw between them.

The prompt

“Which factors, events, groups, or institutions do you think caused Democrats and Republicans to view each other more negatively than they did 50 years ago? Please explain what caused what, who or what you think was responsible, and what effects followed.”

Example 2: Open-Ended Answers as Causal Graphs



Jankin, Liao & Tang (2026), Fig. 4: popular folk theories of the divide, grouped by root cause (% = within-party share)

- Same factors, opposite arrows — only the direction tells the theories apart; the graph gives each person's **root cause**.
- **Validated:** coders agree on which factors each answer names ($\kappa = 0.84$); an LLM with the same codebook matches them ($F1 = 0.92$). **Role of text:** the theory you hold predicts tolerance of anti-democratic action, net of affect and ideology.

Part 2 — How This Course Works

policy, schedule, assessment

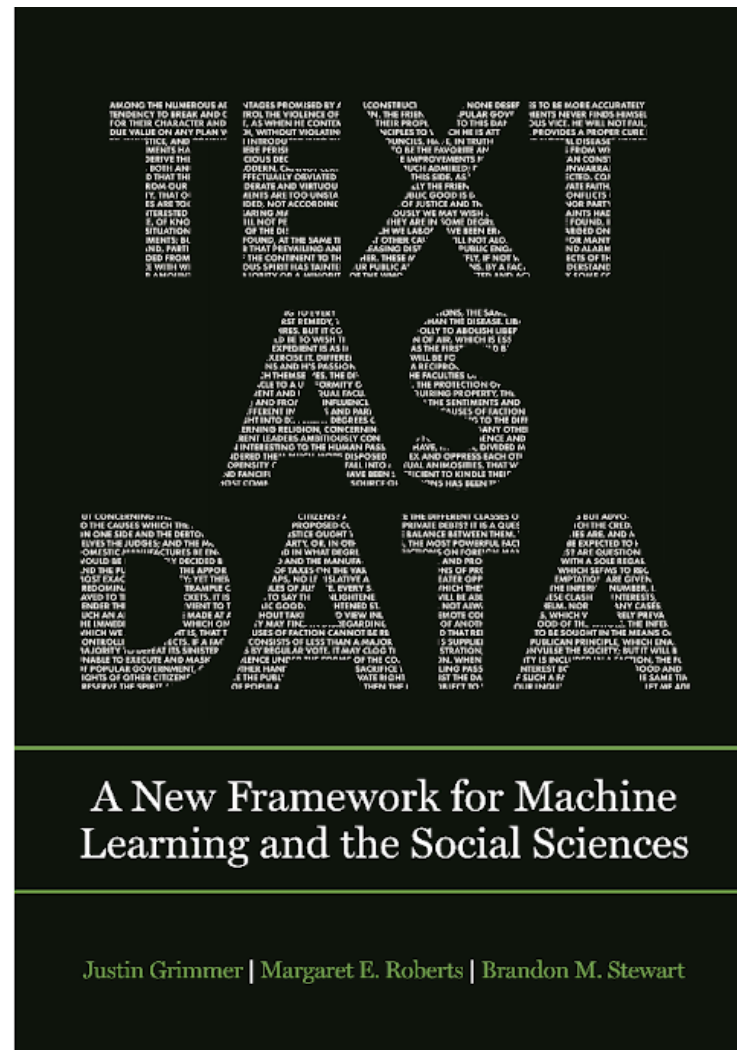
Who This Course Is For

A social science course — **not a technology course.**

This is an introductory course designed for students in the Graduate Institute of National Development and related social science disciplines. No advanced technical training is required. Instead, the course emphasizes how text-as-data and NLP techniques can be applied to answer substantive research questions in political science, political communication, legal studies, sociology, and other social science fields.

- **Want the technical detail?** Model internals, architectures, optimization — this is not that course.
- **The goal:** your own **political science / social science paper**, written with these methods.

Where the Course Material Comes From



The framework behind these lectures

Grimmer, Roberts & Stewart (2022) — *Text as Data: A New Framework for Machine Learning and the Social Sciences*, Princeton University Press

- Our two tasks simplify this book's discovery → measurement → inference framework

You do not need to buy it. This course is **paper-first** — 3–4 core items a week, mostly journal articles and open tutorials. Required book chapters are provided as excerpts.

Also useful: Benoit (2020) · Nielbo et al. (2024) · Watanabe & Müller, *quanteda tutorials* · Jurafsky & Martin, *SLP* (free online)

Course Schedule: 16 Weeks

Week	Date	Topic	Notes
W1	Sep 11	Introduction to Quantitative Text Analysis	
W2	Sep 18	Text Preparation in R and Python: Collaborative Workflow	
W3	Sep 25	Mid-Autumn Festival (No Class)	
W4	Oct 2	Tokenisation, Document-Feature Matrices, Dictionaries, and Sentiment Analysis	
W5	Oct 9	National Day Observed (No Class)	
W6	Oct 16	Human Coding and Document Classification	
W7	Oct 23	Supervised, Unsupervised, and Semi-Supervised Scaling	Individual research abstract due
W8	Oct 30	Topic Models and Their Variants	
W9	Nov 6	Describing and Comparing Texts: Readability, Similarity, and Word Embeddings	
W10	Nov 13	NLP Pipelines and Transformers: Annotation, Classification, and Interpretability	
W11	Nov 20	NTU Sports Day (No Class)	
W12	Nov 27	Large Language Models and Multi-Agent Systems	
W13	Dec 4	New Forms of Textual Data: Multilingual and Multimodal Sources	Midterm research design report due
W14	Dec 11	Poster Workshop: Format, Visual Presentation, and 5-Minute Talk	Poster revision clinic
W15	Dec 18	Final Poster Presentations	5-minute talks; peer feedback
W16	Dec 25	Constitution Day Holiday (No Class)	Manuscript due: TBA; bonus deadline: TBA

Course Schedule: 16 Weeks

The term in three blocks.

W1–W2 • Foundations

Introduction to QTA; text preparation and R/Python collaborative workflows.

W4–W8 • Core methods

Tokenisation and DFMs; dictionaries and sentiment; human coding and classification; scaling; topic models.

W9–W13 • Advanced representations

Readability, similarity and embeddings; NLP and transformers; LLMs; multilingual and multimodal sources.

Dates worth writing down

- **W7 • Oct 23** — individual research abstract due
- **W13 • Dec 4** — midterm research design report due
- **W14 • Dec 11** — poster workshop
- **W15 • Dec 18** — final poster presentations
- **Manuscript and bonus deadlines: TBA**

What You Need Before We Start

This course requires prior experience with R and comfort running some code in a Python environment. No advanced technical training beyond that.

R — most lectures and methods

quanteda, tidytext, text2vec, udpipe, keyATM, spacyr

Python — selectively

transformers, flair (transformer models); gensim, spacy (NLP toolkits); BERTopic (topic models)

You should bring a laptop with a working development environment already set up — most sessions mix a short lecture with coding exercises you run yourself. A detailed setup guide is sent before the first class.

What You Need Before We Start

IDE: RStudio for R, Positron for Python and cross-language work.

You should be able to edit and run code, debug, manage packages, and use the integrated terminal.

R Markdown / Quarto (RStudio) and Jupyter / Quarto notebooks (Positron) are helpful but not required — we cover the practical setup and workflow in Week 2.

New to these IDEs? Complete a basic tutorial before Week 2 — so the term is spent on methods, not on fighting your tools.

Written coursework and posters use **LaTeX in shared Overleaf projects**. Quarto can be used for analysis notebooks. Optional: we may use LM Studio to run LLaMA or Mistral locally, with no API costs.

What You Need Before We Start

GitHub for code; Overleaf for written coursework.

Course materials — lectures, code, data, and practice exercises — are on GitHub. Develop research code in a **private repository**. Submit written coursework through **shared Overleaf projects with instructor editing access**.

No Git experience required. We use **GitHub Desktop** (desktop.github.com) — point-and-click, no command line. RStudio and Positron have built-in Git tools too.

Recommended: apply for the free **GitHub Student Developer Pack** (education.github.com/pack) with your university email. It gives you GitHub Pro, Copilot, and other developer tools at no cost.

Week 1: course rules, expectations, and how materials, coursework, and communication are organized. **Week 2:** GitHub Desktop and IDE setup — cloning, committing, and sharing code.

What You Need Before We Start

This is a 2-credit course. Plan on roughly 3 hours per week outside class.

Activity	Hours / week
Before class — read the assigned text-as-data papers, review the week’s slides	1–2
After class — self-study coding practice and research-project development	1–2
Typical weekly total (outside class)	about 3

On the readings: the assigned papers come from political science and the social sciences and use text-as-data methods on real research problems. We read them for the methods and how they are applied, not for their substantive findings.

Practice and Class Participation

Self-Study Coding Practice

Lab files provide worked examples and exercises for practising course methods.

These practice exercises are ungraded; no lab submission is required.

Class Participation — 20% Active engagement in:

- Discussion of readings, methods, and applications
- Questions and answers on NTU COOL or course-code GitHub Issues
- Peer feedback activities

Written coursework and posters are prepared in LaTeX and submitted through shared Overleaf projects.

Final Poster Presentation

Final Poster Presentation — 25%

Week 15 poster session.

You present a **5-minute overview** of your poster, followed by discussion and peer feedback. You are **expected to remain by your poster throughout the session** to answer questions. Faculty members may be invited to attend.

Posters should follow **PolMeth poster guidelines**

(polmeth.emory.edu/instructions.html): clear visual design emphasizing the research question, data and methods, key findings, and implications.

Week 14 is a preparatory poster workshop — poster format, visual presentation, and the 5-minute talk. It counts toward class participation.

Your Two Papers

Midterm Research Design Report — 20%.

Your own research project or thesis, written as **independent work**. The midterm sets up the study that the final paper completes.

What it must contain

- Abstract
- Introduction: question and motivation
- Literature review situating the study
- Research design: data, methods, strategy

And then

- Preliminary results, or the planned analytical strategy
- Feasibility, expected findings, contribution

Up to 10 pages, no more than 2,500 words. Due: December 4 (Week 13).

Your Two Papers

Final paper — 25%.

Also independent work. Build on the midterm and complete it with the full empirical analysis — stronger design, robust inference, proper validation and sensitivity tests, then findings, limitations, contribution.

- **Main text: up to 4,000 words, excluding references**, written as a **PSRM Research Note**
- **Appendix: no word or page limit** — robustness checks, alternative specifications, validation, annotation protocols, prompts, model details, code documentation
- **A reproducible code repository**, on GitHub or CodeOcean
- Due: TBA

Your Two Papers

Where to start.

The syllabus lists these as suggested topics, if you do not already have a corpus in mind.

Legislatures and parties

- Taiwan legislative behavior: roll call votes, speeches, parliamentary questions
- Legislative speech analysis
- Party manifestos, across countries or over time

Media and rhetoric

- Social media political discourse
- News framing of political issues
- Comparative political rhetoric
- CCP wolf warrior diplomacy and propaganda

Pick something you can still be working on at the Week 15 poster session.

Grading

Component	Weight	Deadline
Research Paper Abstract	10%	Oct 23 (Week 7)
Midterm Research Design Report	20%	Dec 4 (Week 13)
Class Participation	20%	Throughout the semester
Final Poster Presentation	25%	Dec 18 (Week 15), in class
Final Research Paper	25%	TBA
Total	100%	

- **Research abstract** — proposed title and up to **1,000 words**; due **Oct 23**
- **Midterm report → final paper** — develop one original research project
- **Poster presentation** — workshop in W14; presentation in W15
- **Coding practice** — ungraded; no submission
- Submit assessed work via **Overleaf**; up to **15% extra credit**

Bonus: Up to 15% Extra Credit

Two bonuses: a conference acceptance, or an actual journal submission.

Conference acceptance — up to 5%

APSA (early Jan), MPSA (Nov), EPSS (late Dec), COMPTTEXT (early Jan), TPSA (summer), JSQPS (late Oct for Winter, Jul/Aug for Summer); PolMeth — Annual Meeting (Mar), Europe PolMeth (mid-to-late Nov), Asian PolMeth (mid-to-late Oct). Months shown are typical submission deadlines, not meeting dates.

Journal submission — up to 10%

Revise & Resubmit, or major/minor revision: **+10%**. Reject or desk reject: **+5%**. Eligible journals: SSCI-indexed political science or social science journals.

Land both and you can claim both — 15% maximum.

Bonus: Up to 15% Extra Credit

What you have to show, and what makes a submission ineligible.

Proof: the confirmation email showing the submission **and** the decision letter.

Not eligible: a submission that breaches NTU regulations on academic integrity, research ethics, student conduct, authorship, copyright, data protection, or generative-AI use, or the rules of the relevant conference or journal. It may also be handled under the relevant NTU procedures.

Writing an MA thesis? You are encouraged to develop thesis-related submissions with your thesis supervisor. GIND students: a thesis-related co-authored submission must be co-authored with a faculty member affiliated with the Graduate Institute of National Development. For any thesis-related co-authored submission, you must be listed as both **first author** and **corresponding author** to claim the bonus.

Bonus: Up to 15% Extra Credit

Affiliation and geographic identification — required in every bonus submission.

Use the official name **National Taiwan University** in the manuscript text, submission metadata, cover letter, acknowledgements, slides, posters, and any public-facing materials.

Do not alter, omit, or downgrade the university name, and do not use wording that misrepresents or diminishes Taiwan's status. Where a location is required, use “Taipei, Taiwan” or “Taipei, Taiwan (R.O.C.)”.

This does not prohibit evidence-based criticism of Taiwanese politics, institutions, or public policy. It applies to accurate affiliation, geographic description, professional conduct, and non-misleading representation.

If a conference, journal, submission system, or editor asks for wording that changes NTU or Taiwan, you must consult me before proceeding — and you are strongly encouraged to discuss any submission plan with me in advance.

Using Generative AI

This course follows NTU's **Guidelines for the Use of Generative AI and Research Integrity**. AI is a support tool: it does not replace your judgment or your responsibility for what you submit.

1. Permitted uses in your assignments — conceptualisation; initial literature screening, for human verification; supporting writing and editing; audiovisual and interactive components. The argument and the analysis stay your own.

2. Disclose AI use in every submission — research abstract, midterm report, poster, final paper, and bonus submission. State which tool and version, what you asked it to do, how you used the output, and how you checked it. No AI used? Write: “No generative AI tools were used in the preparation of this assignment.”

3. No declaration, no acceptance. An assignment without a declaration is not accepted until a complete one is provided. Misrepresenting AI use, failing to verify AI output, fabricating sources or data through AI, or using AI to bypass a learning goal may be treated as academic misconduct.

Academic Integrity and Deadlines

Integrity

- Acknowledge every source of ideas, text, code, data, figures, tables, and research materials that is not your own
- Plagiarism includes copying or close paraphrasing without citation, reusing your own past work unacknowledged, unauthorized collaboration, fabricated data or references, and AI-generated material without the required disclosure
- This course follows NTU policies: suspected misconduct may receive **a zero on the relevant component**; serious or repeated cases may be referred to the department, college, or university

Deadlines

- Deadlines hold. If you cannot meet one, notify me as early as possible and use the NTU student leave procedure; in urgent cases, email me first
- Approved leave, or an approved extension, does not automatically waive the work; I arrange a reasonable make-up when appropriate
- No approved leave and no prior approval: **absent for missed in-class activities; late work may receive no credit**

Dignity, Respect, and Support

National Taiwan University is committed to a **respectful, inclusive, and safe learning environment**.

You have the right to study free from bullying, harassment, discrimination, sexual harassment, sexual assault, sexual bullying, stalking, and other conduct that undermines dignity or safety.

This course reads politically sensitive material. Disagreement is expected, but it stays evidence-based, professional, and respectful — argue with the evidence, not the person.

If you experience or witness harassment, gender-based misconduct, discrimination, bullying, or a threat to your safety, you are encouraged to seek support. Reports are handled under the relevant NTU procedures.

[You are also welcome to contact me directly](#) — I will help you find the right route.

Dignity, Respect, and Support

Reporting

Gender Equity Education Committee
(性別平等教育委員會) complaint lines: (02)
3366-9607 / (02) 3366-9608, gender@ntu.edu.tw

NTU sexual harassment prevention contact line:
(02) 3366-5903, leader@ntu.edu.tw

Or speak to me first if you prefer.

Support and emergencies

Student Counseling Center: (02) 3366-2181,
my.ntu.edu.tw/counsel — an initial intake
appointment is required; walk-ins welcome in
urgent situations

In-college student support, a counselor based in
your own college (walk-ins welcome):
csw.ntu.edu.tw

Campus Security Center, 24 hours: (02) 3366-9119

If life or physical safety is in danger: **119** (medical)
or **110** (police)

Free national hotlines, 24 hours: Mental Health Care
Line **1925**; Lifeline **1995**

Save the 24-hour numbers in your phone now, before you need them.

Before You Leave: Background Survey

Take five minutes now.

Student Background and Course Expectations

- Where you are with R, Python, and text analysis
- What you want out of the course, and what you are working on
- I use the answers to set the pace of the labs and pick examples that fit your projects

Scan the code and fill it in before you go.



Before Next Week

W2: R, Python, GitHub, and Collaborative Workflow

- Bring a laptop with a **working development environment** — we build on it every week
- Install **RStudio** (R) or **Positron** (Python and cross-language work); a setup guide goes out before class
- Create a free **GitHub account** and install **GitHub Desktop** — no command-line Git required
- **Recommended:** apply for the free **GitHub Student Developer Pack** with your NTU email — GitHub Pro and Copilot at no cost

Reading for W2: *quanteda tutorials*, Ch. 1 · Grimmer, Roberts & Stewart (2022), Ch. 4 (excerpt provided) · Quarto “Get Started” docs · Posit blog, *Privacy and AI Assistants* (skim)

Catching up on W1? Benoit (2020) · Grimmer & Stewart (2013) · Nielbo et al. (2024) · Egami et al. (2022)

References (1/4)

Grimmer & Stewart (2013)

Grimmer, J. & Stewart, B. M. (2013). Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts. *Political Analysis* 21(3): 267–297.

Grimmer, Roberts & Stewart (2022)

Grimmer, J., Roberts, M. E. & Stewart, B. M. (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press.

Jankin, Liao & Tang (2026)

Jankin, S., Liao, Y.-C. & Tang, L. (2026). Folk Theories of Partisan Conflict. Working paper.

Rodríguez, Spirling & Stewart (2023)

Rodríguez, P. L., Spirling, A. & Stewart, B. M. (2023). Embedding Regression: Models for Context-Specific Description and Inference. *American Political Science Review*.

[doi:10.1017/s0003055422001228](https://doi.org/10.1017/s0003055422001228)

References (2/4)

Benoit (2020)

Benoit, K. (2020). Text as Data: An Overview. In L. Curini & R. Franzese (eds.), *The SAGE Handbook of Research Methods in Political Science and International Relations*: 461–497. Sage.

[doi:10.4135/9781526486387.n29](https://doi.org/10.4135/9781526486387.n29)

Egami et al. (2022)

Egami, N., Fong, C. J., Grimmer, J., Roberts, M. E. & Stewart, B. M. (2022). How to Make Causal Inferences Using Texts. *Science Advances* 8(42): eabg2652. [doi:10.1126/sciadv.abg2652](https://doi.org/10.1126/sciadv.abg2652)

Grimmer (2010)

Grimmer, J. (2010). A Bayesian Hierarchical Topic Model for Political Texts: Measuring Expressed Agendas in Senate Press Releases. *Political Analysis* 18(1): 1–35. [doi:10.1093/pan/mpp034](https://doi.org/10.1093/pan/mpp034)

Heseltine & Clemm von Hohenberg (2024)

Heseltine, M. & Clemm von Hohenberg, B. (2024). Large Language Models as a Substitute for Human Experts in Annotating Political Text. *Research & Politics* 11(1).

[doi:10.1177/20531680241236239](https://doi.org/10.1177/20531680241236239)

References (3/4)

Jurafsky & Martin (2026)

Jurafsky, D. & Martin, J. H. (2026). *Speech and Language Processing* (3rd ed., online draft).
<https://web.stanford.edu/~jurafsky/slp3/>

Laver, Benoit & Garry (2003)

Laver, M., Benoit, K. & Garry, J. (2003). Extracting Policy Positions from Political Texts Using Words as Data. *American Political Science Review* 97(2): 311–331.
[doi:10.1017/s0003055403000698](https://doi.org/10.1017/s0003055403000698)

Nielbo et al. (2024)

Nielbo, K. L. et al. (2024). Quantitative Text Analysis. *Nature Reviews Methods Primers* 4(1): 25.
[doi:10.1038/s43586-024-00302-w](https://doi.org/10.1038/s43586-024-00302-w)

Roberts et al. (2014)

Roberts, M. E., Stewart, B. M., Tingley, D. et al. (2014). Structural Topic Models for Open-Ended Survey Responses. *American Journal of Political Science* 58(4): 1064–1082.
[doi:10.1111/ajps.12103](https://doi.org/10.1111/ajps.12103)

References (4/4)

Slapin & Proksch (2008)

Slapin, J. B. & Proksch, S.-O. (2008). A Scaling Model for Estimating Time-Series Party Positions from Texts. *American Journal of Political Science* 52(3): 705–722.

[doi:10.1111/j.1540-5907.2008.00338.x](https://doi.org/10.1111/j.1540-5907.2008.00338.x)

Watanabe & Müller (2023)

Watanabe, K. & Müller, S. (2023). *Quanteda Tutorials*. <https://tutorials.quanteda.io>